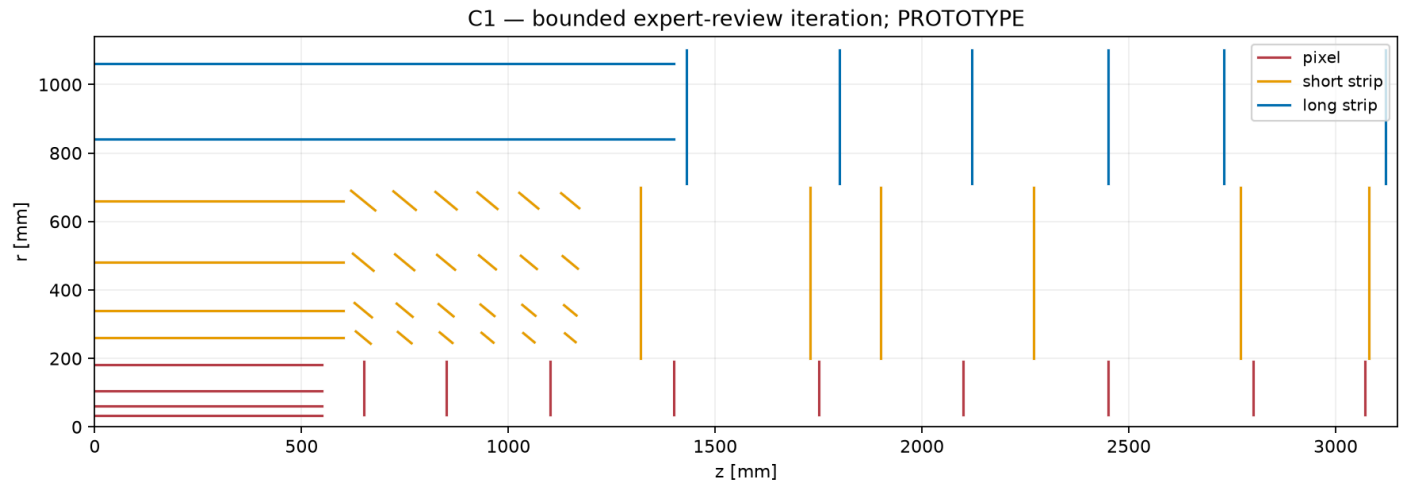


DES-006 C1: expert-review iteration

DRAFT / PROTOTYPE, 2026-09-24. Only A and C1 remain active. Only short-strip barrel ends inclined; outer long strips cylindrical.

Geometry and motivation

Retain the tracker host and $|\eta| < 4$ requirement. Separate active pixel, short-strip and long-strip disk bands with provisional 10 mm gaps. Move the outer long-strip barrels and disks together to recover leverage around $\eta=1.1$. No component or geometry is signed off.



Pixel barrel radii: 34, 60, 106, 182 mm; half-length 550 mm. Short-strip central radii: 260, 340, 480, 660 mm. Long-strip radii: 840, 1060 mm; half-length 1400 mm. First long-strip disk: $z=1430$ mm. Exact signed layers are in DES-006-reviewed-layouts.json.

Disk annuli: pixels 35-190 mm; short strips 200-700 mm; long strips 710-1100 mm. Those active bounds reserve space but do not prove that real supports or services fit. A 25 mm pipe-exclusion plus 5 mm installation fixture is conditional; the 34 mm first radius is not a cleared interface.

C1: strengths, weaknesses and next decision

Demonstrated prototype benefit

At $\eta=1.1$, origin, 3 T, the Gaussian q/p_T uncertainty improves 5.9% at $p_T=1$ GeV and 8.7% at 100 GeV relative to original C1. The generic five-parameter covariance uses straight-reference crossings and leading thin-scatterer noise. It is not a reconstructed resolution or a full material budget.

Ideal silicon areas: pixel 4.612 m²; short strip 37.887 m²; long strip 120.077 m² including both faces. Long-strip pair material is counted once. Earlier controls remain in the original files.

Costs and unresolved weaknesses

The worst finer-grid q/p_T uncertainty ratio is 1.473, so the landmark improvement is not uniform. Narrower disks lose hits in some transitions. The 60 mm second pixel layer trades better high- p_T IP information against about 4% worse central low- p_T d_0 . Far-forward leverage remains limited. These losses prevent adopting the full iterated layout as an optimum.

C1 tested uniform- η row spacing and shifts up to 100 mm. The final gap-constrained choice remains uniform z , zero radial shift, 5 mm margin; constant hit counts were not achieved. Finite-module tiling and services are required before more aggressive row arrangements.

Priorities and human review

Review beam-pipe and service envelopes first; repair lost transition coverage with real module masks; agree beamspot/performance weights; revisit C1 row count and tiling; then run ACTS and later full simulation. Source locators, all attempted scans, adverse cases and individual comment responses are in DES-006-expert-review.md. Final sign-off remains with asalzburger-review.